

NAMOZA HEALTHCARE GROWTH & STRATEGY

Developer Assignment Walkthrough & Presentation Guide

A comprehensive, technical breakdown of GTM schema design, high-performance landing page development, and HubSpot CRM integration logic for **OrthoNow**.

Candidate: Khalid Anwar

Role Applied For: Developer — Position 1 (Client Web + Martech)

Target Account Scenario: OrthoNow (9 orthopaedic clinics)

Submission Repo: github.com/AnwarKhalid00/namoza-assignment

Deployment URL: anwarKhalid00.github.io/namoza-assignment/task2_landing_page.html

Task 01: GTM Event Schema Design

Objective: Establish a tracking framework for OrthoNow to measure patient acquisition funnels, floating widgets, and content engagement without introducing PII risks or heavy script loads.

What was done and why:

- **Implemented 8 critical events:** Developed a robust mapping table (Event Name, Trigger Type, Parameters, and GA4 Destination) covering every interaction: the 3-step booking funnel, click-to-call buttons, WhatsApp chat widget clicks, patient guide downloads, clinic page views, and blog article scroll depths.
- **PII Compliance:** Excluded user-identifiable data (e.g., patient phone numbers or names) from the dataLayer pushes. Instead, these are transferred securely on the server-side to the CRM. GTM only processes non-sensitive parameters like *clinic_location*, *specialty*, and *step_number*.
- **Funnel Tracking Methodology:** Set up structured tracking utilizing custom GTM events (booking_step_complete) to map multi-step form drop-offs natively in GA4 Funnel Exploration.

Recommended Google Ads Conversion:

Event: booking_step_complete where step_number = 3 (Booking Confirmed).

Rationale: Unlike Top-of-Funnel (ToF) interactions like 'Call Now' clicks (which may result in immediate hang-ups) or 'Patient Guide Downloads' (low intent research), a Step 3 completion represents a validated request with phone/name details entered and location/specialty chosen. Feeding this clean high-intent signal back to Google Ads allows Smart Bidding algorithms to optimize directly for actual appointment bookings.

■ **The Interviewer Trap (Multi-Step Form Listening):**

GTM cannot automatically detect steps in a JavaScript-rendered multi-step form. A common amateur mistake is assuming GTM's native Form Submission trigger works here. In the presentation, state clearly: 'I wrote custom dataLayer pushes directly into the front-end JS logic on each step's validation. GTM listens for these pushes.'

Task 02: High-Performance Landing Page Build

Objective: Replace a slow, low-converting (2.1%) WordPress appointment page with a highly responsive, fast page targeting Bengaluru working professionals aged 28–50 experiencing back/knee pain.

What was done and why:

- **Single-File Architecture:** Built the page using pure vanilla HTML, CSS, and JS with zero external dependencies (no frameworks, no external Google Fonts, no external libraries). This minimizes HTTP request overhead and CSS parser blocking.
- **System Font Stack & Inline Assets:** Used a system font stack (*-apple-system, BlinkMacSystemFont, Segoe UI, etc.*) to eliminate font-loading layout shifts (CLS) and render-blocking requests. Icons are inline SVGs to avoid loading image assets.
- **Conversion-Focused UX:** Placed a tight 2-field form (Name + Phone) prominently in the hero section. Included doctor credentials (senior MS Orthopaedic specialists), trust badges (clinic counts, ratings), and a mobile-specific sticky CTA bar at the bottom to maximize conversion rates.
- **Rigorous Input Validation:** Programmed client-side validation specifically for Indian mobile numbers (starts with 6-9, exactly 10 digits) to filter out fake leads before hitting server resources.

Performance Results (Core Web Vitals):

PageSpeed Insights Mobile Score: 100 / 100

- First Contentful Paint: 0.8s
- Largest Contentful Paint: 0.8s
- Total Blocking Time: 0ms
- Cumulative Layout Shift: 0

■ **The Interviewer Trap (PageSpeed & Live Firing):**

All templates fail PageSpeed because of generic images, fonts, and script bloat. Your page achieves 100/100 because of raw file efficiency. In the presentation, you will open Chrome DevTools, show the console tab, fill the form live, and let them see the exact `consultation_form_submitted` event push immediately upon submission.

Task 03: HubSpot CRM & Karix WhatsApp Integration

Objective: Architect a reliable, sub-2-minute SLA pipeline connecting landing page form submissions to HubSpot (lead creation) and Karix (automated WhatsApp confirmation) without losing data.

End-to-End Architecture Overview:

Component	Technology	Function / Role
Front-End Form	Vanilla JS Fetch API	Captures inputs, triggers validation, posts to endpoint.
Middleware Layer	Serverless Function (AWS Lambda / Vercel)	Processes payloads, handles security, routes APIs in parallel.
CRM System	HubSpot API (Search + Create)	Deduplicates via phone Search API; stores lead details.
Notification Engine	Karix WhatsApp API	Sends immediate templated confirmation message.
Fallback / Queue	AWS SQS / BullMQ	Queues retries for WhatsApp/HubSpot API errors.

What was done and why:

- **Bypassed HubSpot Forms API / Native Embeds:** These methods rely on the *email address* for deduplication. Because our form only collects Phone numbers, using native embeds would result in duplicate contact profiles for the same user. A custom serverless function calls HubSpot's Contacts Search API first to lookup the phone number.
- **Parallel Requests for Speed:** Used Promise.all() to fire both the HubSpot update and the Karix WhatsApp API call in parallel, ensuring the WhatsApp confirmation executes well within the 2-minute SLA.
- **Failure Resilience (Fallback Setup):** If the Karix API is down, the request is pushed to a message queue (SQS) with exponential backoff. If all WhatsApp retries fail, it falls back to sending a transactional SMS.

■ **The HubSpot Deduplication Trap:**

Always emphasize this in your talk: 'HubSpot's default behavior is to deduplicate on email, not phone. Since our LP is optimized for conversions and only asks for Name/Phone, we had to build a custom endpoint that queries HubSpot's Search API by phone first. This is a common oversight in Indian healthcare campaigns.'

How to Present in the Interview

This is a 30 to 45-minute presentation with the Co-Founders. You want to present as a business-oriented, technical product engineer, not just a coder. Focus on speed, conversion, and data integrity.

Presentation Outline (Time Allocation):

- 1. Intro & Big Picture (3 mins):** State the objective clearly. You are replacing a slow page and building tracking from scratch to lower the CPA (Cost Per Acquisition) and help performance marketers optimize campaigns.
- 2. Task 1: GTM Event Schema (7 mins):** Walk through the table. Focus on PII exclusion, how custom GTM events (booking_step_complete) are mapped, and how you will build the funnel exploration report in GA4. Explain why you chose Step 3 as the main Google Ads conversion.
- 3. Task 2: Page Live Demo & Performance (10 mins):** Show the page. Explain the conversion architecture (Name/Phone layout, trust signals). Bring up DevTools Console, fill the form, and show the GTM event firing. Show the PageSpeed 100/100 mobile score.
- 4. Task 3: Integration Design (7 mins):** Diagram the Vercel/AWS Lambda middleware approach. Explain why native HubSpot forms were rejected (deduplication issue), and how the Karix backup queue operates.
- 5. Q&A; / Review (10-15 mins):** Be ready for the specific technical questions.

Anticipated Q&A; (Be Ready to Answer):

Q: Why not use a framework like Next.js or React for the landing page?

A: Next.js introduces client-side JS bundle overhead. For a static, 1-form landing page, vanilla HTML/CSS is cleaner, loads instantly, and guarantees a 100/100 PageSpeed score on low-end mobile networks. No server needed.

Q: If the user inputs a typo in their phone number, how does HubSpot handle it?

A: We run strict regex validation client-side before submission. However, for a complete backup, we can hook up a quick OTP or verification API, though that does add form friction.

Q: How will you brief the front-end team to implement Step 2 tracking?

A: I will provide them with the exact dataLayer JSON and instruct them to bind the push to the 'Next' button after Step 2 fields validate successfully.